

3.3.2 Phishing

Lower Bound Computation

$$\begin{bmatrix} \mathbf{x}_{\text{American Samoa}}^{(\text{count})} \\ \mathbf{x}_{\text{American Samoa}}^{(\text{dollars})} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ \$0 & \$0 & \$0 & \$0 & \$0 & \$0 & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} \mathbf{x}_{\text{American Samoa}}^{(\text{count scaled to sample population})} \\ \mathbf{x}_{\text{American Samoa}}^{(\text{dollars scaled to sample population})} \end{bmatrix} = \begin{bmatrix} 3 & \dots & 3 \\ \$0 & \dots & \$0 \end{bmatrix}.$$

$$\bar{X}_{\text{American Samoa}}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{\text{American Samoa}, i}^{(m)},$$

$$m \in \{\text{scaled count, scaled dollars}\}$$

$$\begin{bmatrix} \bar{X}_{\text{American Samoa}}^{(\text{scaled count})} \\ \bar{X}_{\text{American Samoa}}^{(\text{scaled dollars})} \end{bmatrix} = \begin{bmatrix} 3 \\ \$0 \end{bmatrix}.$$

Upper Bound Computation

$$\begin{bmatrix} \mathbf{x}_{\text{California}}^{(\text{count})} \\ \mathbf{x}_{\text{California}}^{(\text{dollars})} \end{bmatrix} = \begin{bmatrix} 2,059 & 3,302 & \dots & 2,041 \\ \$11,073,447 & \$8,864,419 & \dots & \$14,628,440 \end{bmatrix}.$$

$$\begin{bmatrix} \mathbf{x}_{\text{California}}^{(\text{count scaled to sample population})} \\ \mathbf{x}_{\text{California}}^{(\text{dollars scaled to sample population})} \end{bmatrix} = \begin{bmatrix} 9 & 15 & \dots & 15 \\ \$48,402 & \$40,268 & \dots & \$64,505 \end{bmatrix}.$$

$$\bar{X}_{\text{California}}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{\text{California}, i}^{(m)},$$

$$m \in \{\text{scaled count, scaled dollars}\}$$

$$\begin{bmatrix} \bar{X}_{\text{California}}^{(\text{scaled count})} \\ \bar{X}_{\text{California}}^{(\text{scaled dollars})} \end{bmatrix} = \begin{bmatrix} 12 \\ \$40,268 \end{bmatrix}.$$

Monte Carlo Simulations for Likelihood and Impact

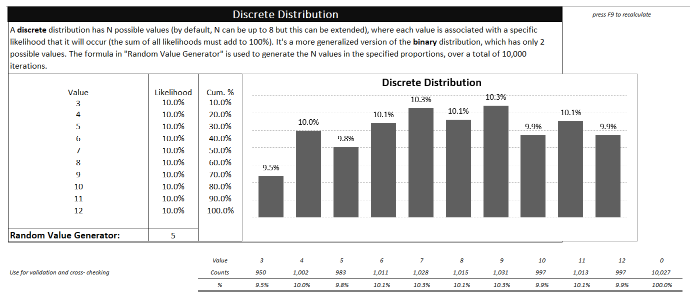


Fig. 3.22: We ran a Monte Carlo simulation of 10,000 iterations with a discrete distribution to spread random instances between upper and lower bounds taken from the means of seven year samples. We bounded between 3 and 12 possible phishing events over a year.[74] [75]

When we computed the upper and lower bounds, we recognized that a normal distribution would not fit well for forecasting likelihood. We took the mean of the sample from American Samoa for the lower bound (3); we took the mean from the sample from California for the upper bound (12); we had a separation of 9 units. Therefore, we switched to a discrete model with all 10 options sharing an equal likelihood.

Table 3.4: Monte Carlo Discrete Simulation Values for Phishing

Events	3	4	5	6	7
Likelihood	9.5%	10.0%	9.8%	10.1%	10.3%

Events	8	9	10	11	12
Likelihood	10.1%	10.3%	9.9%	10.1%	9.9%

A Monte Carlo simulation was run using a discrete distribution. We bounded the simulation between 3 and 12 events. We assigned a 10% likelihood to each bin. The resulting distribution showed the effect of randomness on each possible outcome.

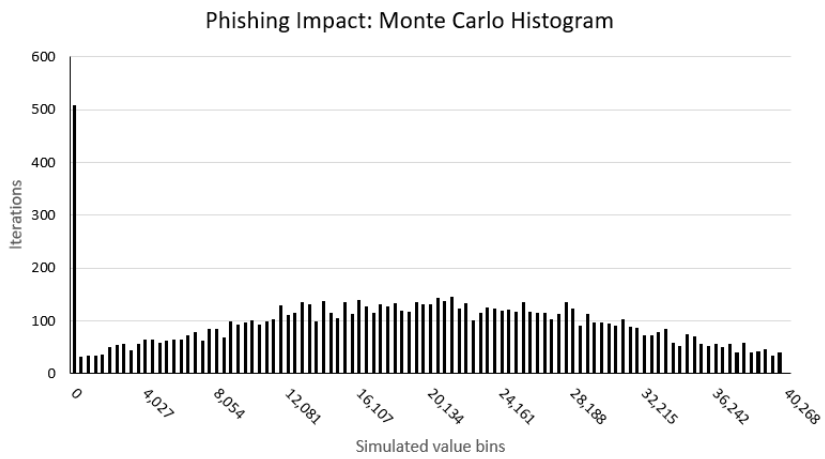
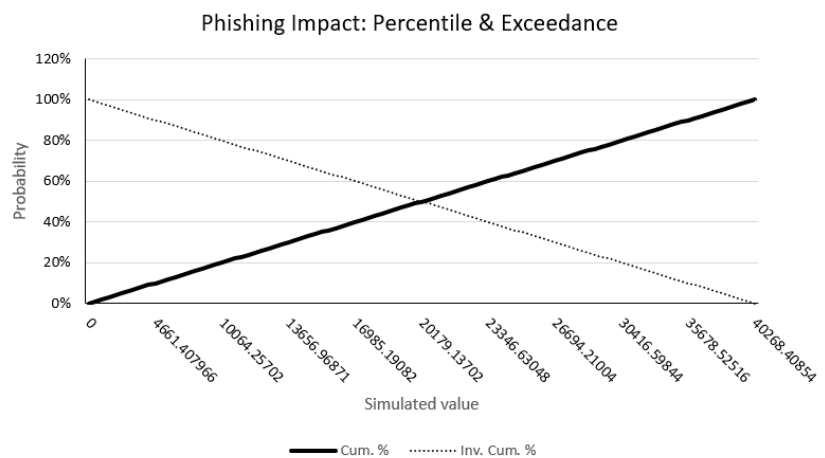


Fig. 3.23: We ran a Monte Carlo simulation of 10,000 iterations with a normal distribution to spread random instances between bounds taken from means of seven year samples. We set a lower bound that the events would cost at \$0. We set an upper bound that the events would cost \$40,268. [76] [77]



$$CI_{80\%}(\theta) = [\$4,661.40, \$35,678.52]$$

Fig. 3.24: A Monte Carlo simulation with a normal distribution suggests that there is a 90% probability that phishing events will cost \$4,661.40 or more in a year. There is a 50% likelihood that they will cost \$20,179.13 or more in a year. There is a 10% chance that they will cost \$35,678.52 or more in a year. [78] [79]